

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 7

Date: Oct 5, 2023

1. In the circuit shown in Figure 1, find the minimum input voltage V_S required to turn on the diode with a cut in voltage of 0.7 V.

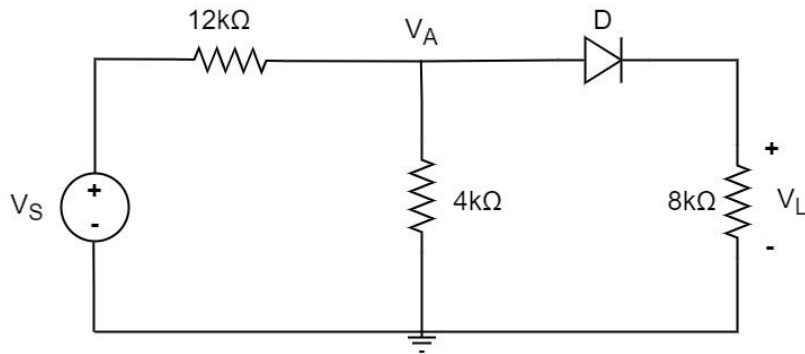


Figure 1

Solution:

$$\frac{V_S - V_A}{12} - \frac{V_A}{4} = \frac{V_A - 0.7}{8}$$

$$\frac{V_S}{12} = V_A \left(\frac{1}{12} + \frac{1}{4} + \frac{1}{8} \right) - \frac{0.7}{8}$$

V_S will be minimum if $V_A = 0.7\text{ V}$ so that diode D just conducts.

So,

$$\frac{V_S}{12} = 0.7 \left(\frac{1}{12} + \frac{1}{4} + \frac{1}{8} \right) - \frac{0.7}{8}$$

$$V_S = 0.7 * 12 \left(\frac{1}{12} + \frac{1}{4} \right)$$

$$V_S = 0.7 * 4$$

$$V_S = 2.8\text{V}$$

2. What will be the relationship between V_o and V_i for the circuit in Figure 2 when the diode D has cut in voltage of 0.7 V.

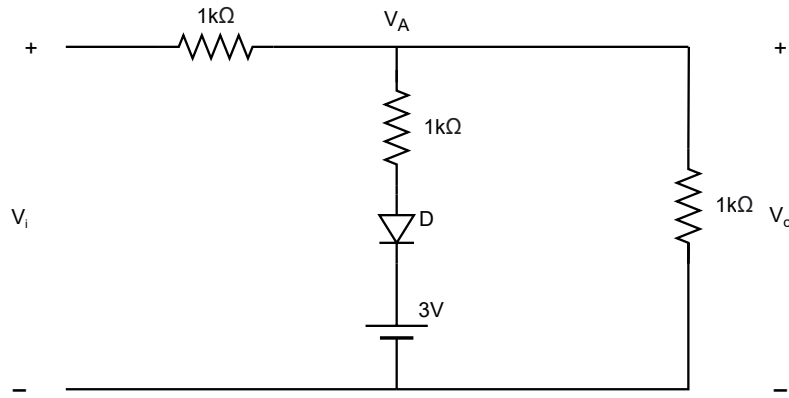


Figure 2

Solution:

$$V_i - V_A = V_A - 3.7 + V_A$$

$$V_i + 3.7 = 3V_A$$

$$V_A = \frac{1}{3}(V_i + 3.7)$$

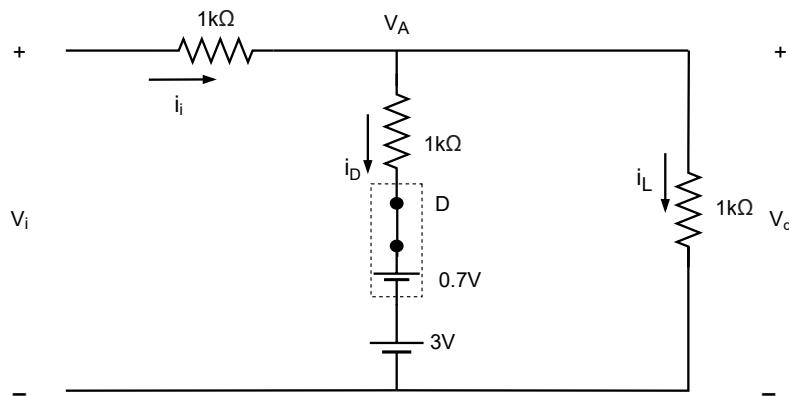


Figure 3

For the diode D to be On, we require

$$V_A \geq 3.7$$

$$\Rightarrow \frac{1}{3}(V_i + 3.7) \geq 3.7$$

$$V_i \geq 7.4V$$

$$V_o = V_A$$

$$V_o = \frac{1}{3}(V_i + 3.7) \quad \text{for} \quad V_i \geq 7.4V$$

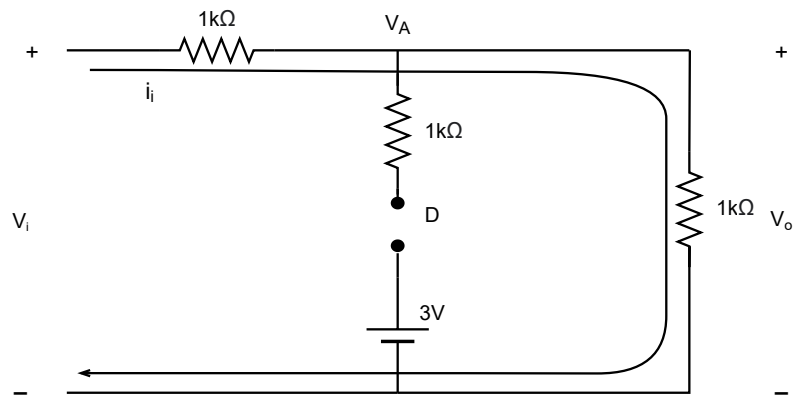


Figure 4

For $V_i \leq 7.4V$, diode D is reverse biased; the transfer function is given by

$$V_o = \frac{V_i}{2} \quad \text{for } V_i \leq 7.4V$$

Hence,

$$V_o = \begin{cases} \frac{V_i}{2} & V_i \leq 7.4V \\ \frac{1}{3}(V_i + 3.7) & V_i \geq 7.4V \end{cases}$$

3. Find $v(t)$ and $i(t)$ in the circuit shown in Fig. 5.

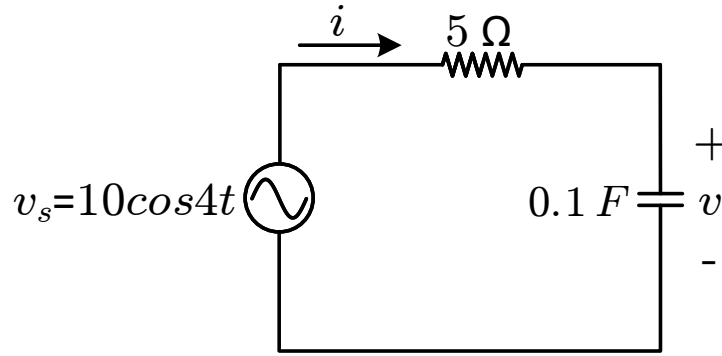


Figure 5

Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\bar{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\bar{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \Omega.$$

Hence the current

$$\begin{aligned} \bar{I} &= \frac{\bar{V}_s}{\bar{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A.} \end{aligned}$$

The voltage across the capacitor is

$$\bar{V} = \bar{I} \cdot Z_c = \frac{\bar{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \cdot 0.1} = \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V.}$$

Converting $\bar{I}$ and $\bar{V}$ into the time domain, we get

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

4. Find the circuit's input impedance in Fig. 6. Assume that the circuit operates at $\omega = 50 \text{ rad/s}$.

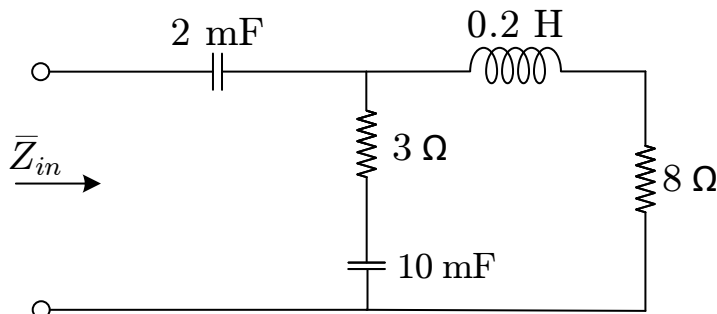


Figure 6

Solution:

Let,

$\bar{Z}_1$ = Impedance of the 2 mF capacitor

$\bar{Z}_2$ = Impedance of the 3 Ω resistor in series with the 10 mF capacitor

$\bar{Z}_3$ = Impedance of the 0.2 H inductor in series with the 8 Ω resistor

Then,

$$\bar{Z}_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega,$$

$$\bar{Z}_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = 3 - j2 \Omega,$$

$$\bar{Z}_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega.$$

The input impedance, $\bar{Z}_{in}$ is given by

$$\begin{aligned}\bar{Z}_{in} &= \bar{Z}_1 + \bar{Z}_2 || \bar{Z}_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(8 + j10)}{11^2 + 8^2} = -j10 + 3.22 - j11.07 \Omega.\end{aligned}$$

Thus,

$$\bar{Z}_{in} = 3.22 - j11.07 \Omega.$$

5. Determine $v_o(t)$ in the circuit of Fig 7.

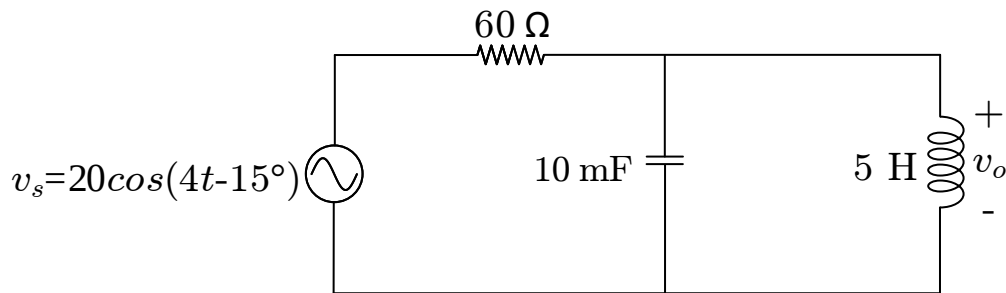


Figure 7

Solution:

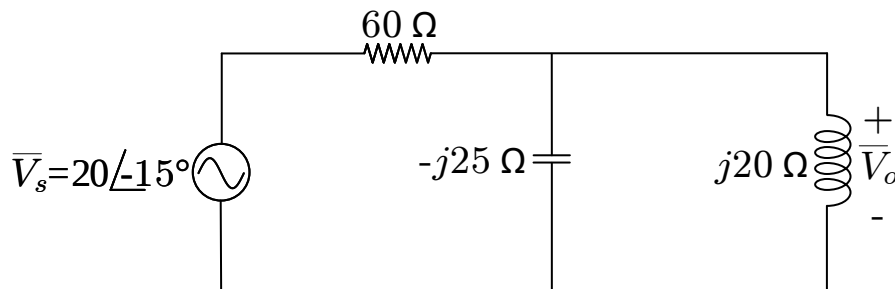


Figure 8

To do the analysis in the frequency domain, we must first transform the time domain circuit in Fig 7 to the phasor domain equivalent in Fig 8. The transformation produces

$$v_s = 20 \cos(4t - 15^\circ) \Rightarrow \bar{V}_s = 20 \angle -15^\circ \text{ V}, \quad \omega = 4$$

$$10 \text{ mF} \Rightarrow \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25 \Omega$$

$$5 \text{ H} \Rightarrow j\omega L = j4 \times 5 = j20 \Omega$$

Let,

$\bar{Z}_1$ = Impedance of the 60 Ω resistor

$\bar{Z}_2$ = Impedance of the parallel combination of the
10 mF capacitor and the 5 H inductor

Then $\overline{Z}_1 = 60\Omega$ and

$$\overline{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \Omega$$

By the voltage-division principle,

$$\begin{aligned}\overline{V}_o &= \frac{\overline{Z}_2}{\overline{Z}_1 + \overline{Z}_2} \overline{V}_s = \frac{j100}{60 + j100} (20 \angle -15^\circ) \\ &= (0.8575 \angle 30.96^\circ) (20 \angle -15^\circ) = 17.15 \angle 15.96^\circ \text{ V}\end{aligned}$$

We convert this to the time domain and obtain

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \text{ V}$$